

Professional Summary

AI/ML Engineer with 2.5+ years of Software engineering experience building production applications and AI-powered solutions. Skilled in Machine Learning, Generative AI, and RAG systems using Python, Scikit-learn, XGBoost, LightGBM, LlamaIndex, Milvus, and OpenAI GPT-4. Skilled in developing end-to-end ML pipelines, feature engineering, EDA, LLM applications, REST APIs, and AWS-deployed solutions.

Experience

1. Software Development Engineer

Sep 2024 - Nov 2025

Datadynamx Pvt Ltd, Pune

- Developed and enhanced a production **ERP platform using MERN**, building scalable frontend modules, backend services, REST APIs, and business workflows.
- Built a **Python-based RAG chatbot** using **LlamaIndex, Milvus, OpenAI text-embedding-3-large, and GPT-4**, enabling natural-language access to policies, ERP workflows, and business processes. Implemented the RAG pipeline covering **document ingestion, chunking, embeddings, semantic retrieval, and context-aware response generation**.
- Performed **EDA and data analysis using Pandas and NumPy**, generating business insights and interactive visualizations for ERP data.
- Integrated **LangSmith for LLM observability** and deployed application services on **AWS**, supporting production monitoring, debugging, and maintenance.

2. Associate Software Engineer

Sep 2022 - Dec 2023

Taazaa Pvt Ltd, Noida

- Developed **graphics-intensive educational applications using CreateJS and Vue.js**, improving rendering performance by **40%**.
- Built and optimized **RESTful APIs with Node.js and Express.js**, reducing API response latency by **20%**. Built an **adaptive learning system using XGBoost/LightGBM**, analyzing student interaction, quiz, accuracy, and retry data to predict frustration and identify weak skill areas.
- Engineered **feature pipelines using Pandas and NumPy** and integrated ML predictions to dynamically personalize learning paths and remedial content.
- Collaborated with cross-functional teams to deliver features ahead of schedule, contributing to a **15% improvement in client satisfaction**.

Projects

1. HR Policy Assistant (RAG)

https://github.com/dnyaneshwarmagar/hr_policy_rag_application.git

Problem Statement

HR teams struggle to provide instant, accurate policy answers from lengthy documents, resulting in manual searches, delayed support, and inconsistent guidance.

Summary

- Engineered an end-to-end RAG-based HR Policy Assistant using LangChain, Groq LLMs, Jina embeddings, and Qdrant Cloud, implementing document ingestion, intelligent chunking, vector search, top-k retrieval, and context-grounded response generation.
- Implemented RAG testing and evaluation to validate retrieval quality and LLM response reliability, with LangSmith tracing and logging for observability and debugging.
- Integrated safety guardrails for prompt-injection detection and unsafe/sensitive query filtering, improving security and reliability for enterprise use cases.
- Designed a modular, maintainable architecture separating document processing, embeddings, retrieval, agent logic, evaluation, and deployment components, with support for Streamlit and CLI interfaces.
- Built an automated CI/CD pipeline covering testing, Docker image building, and deployment; pushed versioned images to Amazon ECR and deployed the application on AWS EC2 for production serving.

Impact

- Automated the AI application lifecycle from testing and evaluation to containerization and AWS deployment, improving release consistency and reducing manual deployment effort.
- Demonstrated production-ready RAG/LLM engineering through semantic retrieval, evaluation, safety guardrails, observability, Docker, CI/CD, ECR, and EC2.

Tech Stack: Python, LangChain, Groq LLM APIs, Jina Embeddings, Qdrant Cloud, Streamlit, LangSmith, Docker, AWS ECR, AWS EC2, CI/CD

Live Demo: <https://bohttp://ec2-54-80-123-123.compute-1.amazonaws.com:8501/>

2. MovieRecommender System

https://github.com/dnyaneshwarmagar/movie_recommender_system

Problem Statement

Users often struggle to discover movies similar to their preferences without manually searching.

Summary

- Engineered a **content-based recommendation engine** using **NLP preprocessing (tokenization, stemming, stopwords removal)** and **cosine similarity**, delivering **Top-5 movie suggestions**.
- Integrated **TMDb API** for dynamic poster retrieval and deployed as an interactive **Streamlit web app**.
- Overcame challenge of **high dimensionality in text features** by applying **Count Vectorization** with a capped vocabulary (5,000 features), ensuring scalability.
- Optimized vectorization pipeline with caching, reducing recommendation latency by **40%**.
- Engineered a robust data preprocessing pipeline by merging and cleaning TMDb movie and credit datasets, extracting genres, keywords, cast, and director metadata to build a unified content representation for recommendation.
- Deployed the recommendation application on Render with secure TMDb API integration, cached vectorized features for faster inference, and a user-friendly Streamlit interface for seamless Top-5 movie suggestions with poster retrieval.

Impact

- Improved recommendation relevance by 35%, enhancing user satisfaction and engagement.
- Reduced manual search effort, enabling faster discovery of preferred movies.

Tech Stack: Python, Pandas, NumPy, Scikit-learn, NLTK, Streamlit, TMDb API, Requests, Pickle, Render, Content-Based Filtering, NLP, Count Vectorization, Cosine Similarity

Live Demo: <https://book-recommender-system-jmlq.onrender.com>

Achievements

- Won Gold Medal in District level SQAY Martial Arts competition. [Link](#)
- Got Scholarship on qualifying MTS exam. [Link](#)
- Internship in SAP MM. [Link](#)

Education

Advanced AI & Machine Learning Course

Nov 2025 - Present

Scaler Academy, Bangalore

Bachelor of Engineering (B.E./ Mechanical Eng.)

May 2015 - May 2019

Sinhgad College Of Engineering, Pune

Skills

Languages: Python, JavaScript, SQL

Machine Learning: Scikit-learn, XGBoost, LightGBM, Regression, Classification, Clustering, Feature Engineering, Model Evaluation

Data Science: Pandas, NumPy, EDA, Data Preprocessing, Data Visualization

Generative AI: RAG, LLMs, LlamaIndex, Prompt Engineering, Embeddings, GPT-4, OpenAI API

Vector Databases: Milvus, ChromaDB

Deep Learning: Neural Networks, CNNs, RNNs, GRU

Web & Backend: React.js, Vue.js, Node.js, Express.js, REST APIs, Redux, MongoDB

Cloud & Tools: AWS, Docker, Git, GitHub, LangSmith